

Neural Networks and Deep Learning Lab-6

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1 LAB-6: Recurrent Neural Networks

1.1 Problem 1: RNN for Next Token Prediction

1.1.1 Objective

- To understand sequence modeling using a vanilla Recurrent Neural Network.
- To analyze the learning behavior and limitations of vanilla RNN.

1.1.2 Hypothesis

If trained on structured sequences of the form $a^k b^k$ followed by EOS, the RNN should learn the dependency between tokens and predict the next token correctly. Gradient clipping is expected to stabilize training by preventing exploding gradients.

1.1.3 Experimental Description

- Synthetic sequences of the form $a^k b^k$ followed by EOS were generated.
- Tokens were converted to one-hot vectors using a vocabulary mapping.
- A vanilla RNN with tanh activation was implemented.
- Cross-entropy loss was used for next-token prediction.
- Training was performed both with and without gradient clipping.
- Loss vs. epochs was recorded and sequence accuracy was computed.

1.1.4 Observations

1. The RNN successfully learned the dependency between tokens ‘a’ and ‘b’.
2. Training loss decreased steadily across epochs.
3. Training without gradient clipping showed higher fluctuations which can be observed in the plot.
4. Training with gradient clipping produced stable convergence.

1.1.5 Inference and Conclusion

Learning Behavior of RNN The vanilla RNN was able to learn the sequential dependency between tokens ‘a’ and ‘b’. The decreasing loss and correct sequence generation demonstrate that the model captured the underlying pattern.

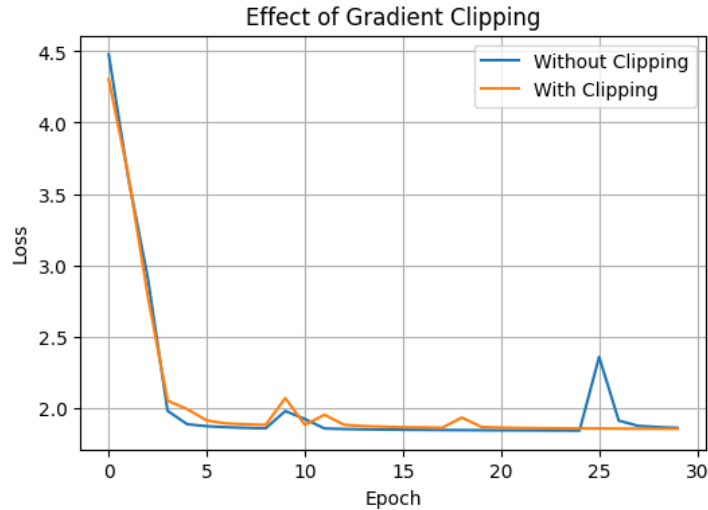


Figure 1: Training loss comparison with and without gradient clipping

Effect of Gradient Clipping Gradient clipping constrained the magnitude of gradients during backpropagation, resulting in more stable training. Without clipping, training exhibited larger fluctuations, indicating the risk of exploding gradients.

Importance of Sequential Memory The hidden state of the RNN retained information from previous time steps, enabling the model to remember the number of preceding ‘a’ tokens before transitioning to ‘b’ and finally predicting EOS. This confirms the importance of sequential memory in recurrent networks.

Limitations of Vanilla RNN Despite learning the basic pattern, the vanilla RNN showed limitations such as slow convergence and weaker confidence in EOS prediction. These issues arise due to limited long-term memory capacity, motivating the use of advanced architectures like LSTM and GRU.

1.2 Problem 2: RNN for Time Series Prediction (Sine Wave)

1.2.1 Objective

- To apply RNN for time-series forecasting.
- To predict the 51st value of a sine wave using previous 50 values.

1.2.2 Hypothesis

Since the sine wave is periodic, an RNN with sequential memory should learn the temporal pattern and accurately predict the next value in the sequence.

1.2.3 Experimental Description

- A sine wave of length 200 was generated.
- Sliding windows of length 50 were used as inputs, with the 51st value as target.
- A vanilla RNN with linear output layer was implemented for regression.

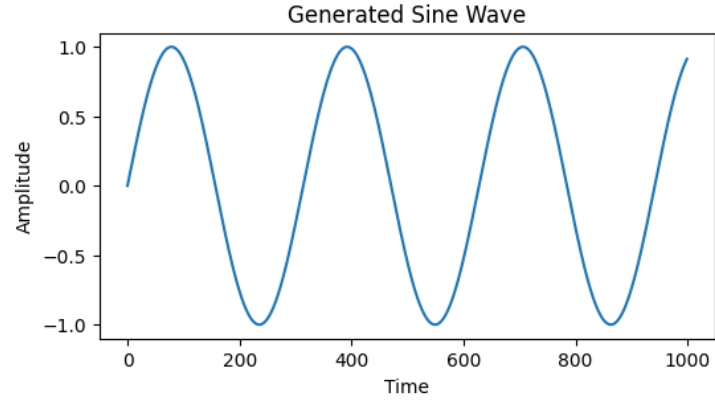


Figure 2: Training loss comparison with and without gradient clipping

- Mean Squared Error (MSE) was used as the loss function.
- The model was trained using Backpropagation Through Time.
- Predicted values were compared against actual sine values.

1.2.4 Observations

1. The RNN successfully captured the periodic behavior of the sine wave.
2. Prediction error decreased steadily with training epochs.
3. The predicted curve closely followed the actual sine wave, demonstrating the regression capability of RNN.

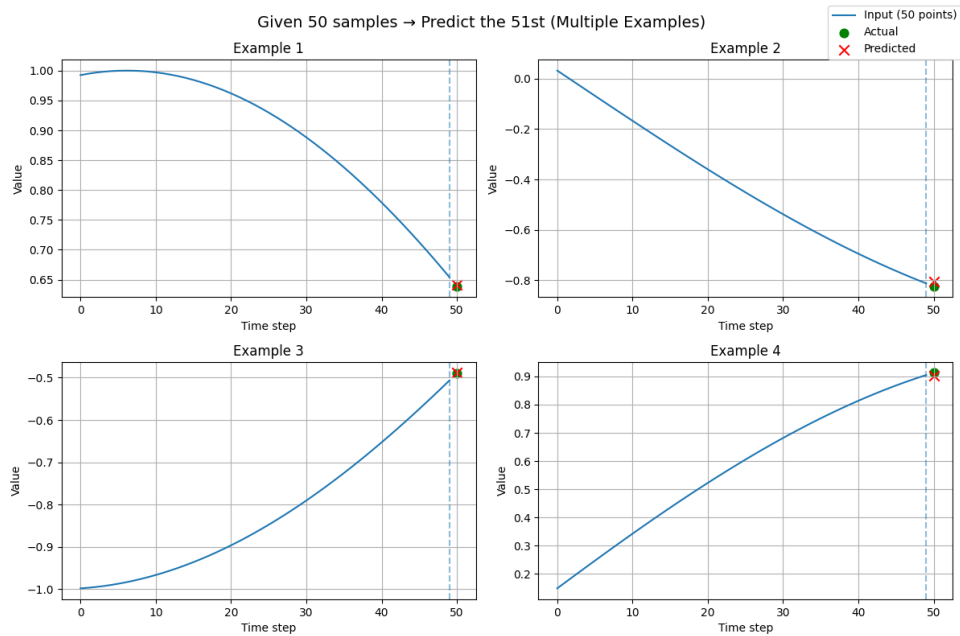


Figure 3: Actual vs Predicted sine wave values

1.2.5 Conclusion

The vanilla RNN effectively modeled temporal dependencies in the sine wave data and was able to predict the next time-step value with low error. This experiment demonstrates the applicability of RNNs for time-series regression tasks. However, for longer and more complex sequences, more advanced recurrent architectures may provide improved performance and stability.

Overall Conclusion: The experiments confirm that vanilla RNNs can learn short-term sequential patterns in both symbolic and continuous domains. However, their limitations in handling long-term dependencies highlight the need for improved architectures such as LSTM and GRU for more complex sequence modeling tasks.